Nonparametric Predictive Regression with Exogenous Shocks: A Sieve Empirical Likelihood Approach (In progress)

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1 Introduction and Data-Generating Process

We consider the problem of testing return predictability in the presence of highly persistent predictors and exogenous covariates (such as weather shocks). The data-generating process (DGP) is specified by the following system of equations:

1. **Predictive Regression:**

$$Y_t = m(W_{t-1}) + \beta X_{t-1} + U_t \tag{1}$$

$$X_t = \theta + \rho X_{t-1} + \varepsilon_t \tag{2}$$

2. **Endogeneity:**

$$U_t = \phi \varepsilon_t + z_t \quad \text{with } E[z_t \mid \varepsilon_t] = 0 \tag{3}$$

where Y_t represents the asset returns, W_{t-1} is a stationary exogenous weather covariate, and X_{t-1} is a persistent financial predictor. The key timing restriction is daily contemporaneous orthogonality: realized local weather shocks W_{t-1} are orthogonal to the predetermined price predictor X_{t-1} at the sampling frequency used for inference. This restriction does not rule out weather effects on subsequent returns through $m(W_{t-1})$, nor does it rule out price adjustment in future values of X_t.

Motivated by Perron (1989), our goal is to investigate return predictability while avoiding size distortions or false inferences caused by near unit roots in the predictor dynamics (i.e., $\rho \approx 1$). This is achieved by testing the null hypothesis of no predictability:

$$\begin{aligned} H_0 : &\quad \beta = 0 \\ H_1 : &\quad \beta \neq 0 \end{aligned} \tag{4}$$

By extending the empirical likelihood (EL) framework of Cai et al. (2026), we propose a two-stage Orthogonalized Sieve-EL framework that achieves a standard χ^2_1 limiting distribution under the null.

2 Empirical Motivation: Weather Shocks and Cocoa Futures

We motivate our framework with the 2023–2024 agricultural commodity shocks. Let Y_t denote excess returns on cocoa futures, W_{t-1} an exogenous local weather variable such as precipitation or average temperature, and X_{t-1} a persistent financial predictor, such as the lagged price.

During 2023–2024, extreme weather conditions, including severe droughts in West Africa, led to supply deficits and a substantial increase in cocoa prices. This context illustrates the key features of our model:

1. **Exogeneity of Weather Shocks:** Local weather variables (W_{t-1}) are realized shocks at the daily sampling frequency. They are contemporaneously orthogonal to predetermined financial variables (X_{t-1}) such as trading volume or lagged prices, while still being allowed to affect subsequent returns and future prices. This timing restriction is formalized in Assumption 2.

2. **Nonlinear Weather Impact:** The effect of temperature and rainfall on crop yield is highly nonlinear, with extreme weather deviations causing disproportionate impacts on supply. A nonparametric formulation $m(W_{t-1})$ is necessary to capture this threshold behavior, which we approximate using a sieve basis.

3. **Predictor Persistence:** Financial predictors (X_{t-1}) often exhibit near unit root behavior ($\rho \approx 1$). Standard OLS inference on β with $m(W_{t-1})$ and persistent X_{t-1} leads to size distortions.

4. **Orthogonalized Sieve-EL Application:** Our framework first filters the nonlinear weather impact $m(W_{t-1})$ via sieve projection. The contemporaneous orthogonality of W_{t-1} and the predictor weight $w(X_{t-1})$ ensures that this projection does not remove the persistent predictor signal. Centering the predictor's weights against the sieve basis purges the nonparametric estimation error from the EL moment condition, allowing for standard χ^2_1 inference on β.

3 The Orthogonalized Sieve-EL Framework

We handle the unknown nonparametric curve $m(W_{t-1})$ by approximating it using a sieve basis of dimension K (e.g., polynomials or splines). Let $\boldsymbol{P}_K(W_{t-1}) = (p_1(W_{t-1}), \ldots, p_K(W_{t-1}))'$ be the $K \times 1$ vector of these basis functions.

3.1 The Step-by-Step Algorithm

The estimation and inference procedure is implemented sequentially via the following three steps, detailing the explicit matrix algebra that guarantees robustness.

Step 1: Profiling out the Weather Curve

For each candidate parameter β under the null hypothesis, we profile out the nonparametric weather curve $m(W_{t-1})$. Let $\boldsymbol{Y} = (Y_1, \ldots, Y_T)'$ and $\boldsymbol{X}_{lag} = (X_0, \ldots, X_{T-1})'$. Let $\boldsymbol{P}$ be the $T \times K$ matrix whose t-th row is $\boldsymbol{P}_K(W_{t-1})'$. The profile sieve coefficients for the weather curve are:

$$\widehat{\boldsymbol{c}}_K(\beta) = (\boldsymbol{P}'\boldsymbol{P})^{-1}\boldsymbol{P}'(\boldsymbol{Y} - \beta \boldsymbol{X}_{lag}) \tag{5}$$

The residualized dependent variable (the adjusted return) is defined using the sieve annihilator matrix $\boldsymbol{M}_K = \boldsymbol{I}_T - \boldsymbol{P}(\boldsymbol{P}'\boldsymbol{P})^{-1}\boldsymbol{P}'$:

$$\widehat{\boldsymbol{u}}(\beta) = \boldsymbol{M}_K(\boldsymbol{Y} - \beta \boldsymbol{X}_{lag}) \tag{6}$$

Under the null hypothesis $\beta = \beta_0$, substituting the true DGP $\boldsymbol{Y} = \boldsymbol{P}\boldsymbol{c} + \boldsymbol{r} + \beta_0 \boldsymbol{X}_{lag} + \boldsymbol{U}$ (where $\boldsymbol{r}$ is the approximation error) yields:

$$\widehat{\boldsymbol{u}}(\beta_0) = \boldsymbol{M}_K(\boldsymbol{P}\boldsymbol{c} + \boldsymbol{r} + \boldsymbol{U}) = \boldsymbol{M}_K(\boldsymbol{r} + \boldsymbol{U}) \tag{7}$$

Step 2: The Sieve Weight Projection

In standard Empirical Likelihood, the nonparametric estimation error would severely distort the asymptotic distribution. To neutralize this, we construct an orthogonalized weight function. Let $\boldsymbol{w} = (w_1, \ldots, w_T)'$ where $w_t = w(X_{t-1})$ is the raw weight (e.g., $\tanh(X_{t-1}/c_w)$). We project the weights onto the same sieve basis and define the orthogonalized weight $\boldsymbol{w}^c$ as the residual:

$$\boldsymbol{w}^c = \boldsymbol{M}_K \boldsymbol{w} \tag{8}$$

Let $w_t^c = (\boldsymbol{w}^c)_t$. By the properties of orthogonal projection, the centered weights are perfectly orthogonal to the sieve basis in-sample:

$$\sum_{t=1}^{T} \boldsymbol{P}_K(W_{t-1}) w_t^c \equiv \boldsymbol{0} \tag{9}$$

Step 3: Algebraic Elimination in the EL Moment Condition

For testing $H_0 : \beta = \beta_0$, the EL moment condition is evaluated as:

$$\widehat{Z}_t(\beta_0) = \widehat{u}_t(\beta_0) w_t^c \tag{10}$$

Summing the moment conditions across the sample to evaluate the numerator of the EL statistic:

$$\sum_{t=1}^{T} \widehat{Z}_t(\beta_0) = \widehat{\boldsymbol{u}}(\beta_0)' \boldsymbol{w}^c = (\boldsymbol{r} + \boldsymbol{U})' \boldsymbol{M}_K \boldsymbol{M}_K \boldsymbol{w} = (\boldsymbol{r} + \boldsymbol{U})' \boldsymbol{w}^c \tag{11}$$

The multi-dimensional Sieve estimation error is thereby perfectly annihilated algebraically. We are left with:

$$\frac{1}{\sqrt{T}} \sum_{t=1}^{T} \widehat{Z}_t(\beta_0) = \frac{1}{\sqrt{T}} \sum_{t=1}^{T} U_t w_t^c + \underbrace{\frac{1}{\sqrt{T}} \sum_{t=1}^{T} r_t w_t^c}_{= o_p(1)} \tag{12}$$

Because the condition is structurally stripped of estimation variance, the sample variance of $\widehat{Z}_t(\beta_0)$ accurately reflects the pure innovation variance. As a result, the Empirical Likelihood ratio statistic (where $\widehat{\lambda}$ is the Lagrange multiplier) achieves the standard chi-squared limit:

$$\ell_T(\beta_0) = 2 \sum_{t=1}^{T} \log(1 + \widehat{\lambda} \widehat{Z}_t(\beta_0)) \xrightarrow{d} \chi_1^2 \tag{13}$$

3.2 Comparison with Cai et al. (2026)

Our framework differs from the approach of Cai et al. (2026) in two fundamental aspects:

- **Nuisance Dimensionality**: Cai et al. (2026) handle a fixed scalar intercept α, whereas our model contains an infinite-dimensional nuisance function $m(W_{t-1})$, which requires a sieve approximation of dimension $K \to \infty$.
- **Orthogonalization Space**: The true generalization is that the demeaning against a constant 1 is replaced by residualizing against the Sieve basis $\boldsymbol{P}_K(W)$. In Cai et al. (2026), the weight centering $w^c(x) = w(x) - \frac{1}{T} \sum w(X_s)$ represents a projection onto the constant subspace span$\{\mathbf{1}\}$. In contrast, our orthogonalized weight $\boldsymbol{w}^c$ is projected onto the K-dimensional sieve space span$\{\boldsymbol{P}_K(W_{t-1})\}$. We keep the AR(1) equation as the persistence DGP, but we do not use AR(1) OLS residuals in the EL statistic.

4 The Growth Rate of the Sieve Dimension K

To establish the asymptotic validity of our test statistic, we must place bounds on the growth rate of the tuning parameter K relative to the sample size T.

4.1 Lower Bound (Bias Elimination)

Assume the true weather curve $m(W)$ has smoothness s (i.e., is s times continuously differentiable). Sieve theory implies that the approximation error r_t satisfies $\sup |r_t| = O_p(K^{-s})$. To ensure that the approximation bias does not distort the asymptotic distribution, we require $\sqrt{T} \sum_{t=1}^{T} r_t w_t^c = o_p(1)$, which translates to:

$$\sqrt{T} K^{-s} \to 0 \quad \Longrightarrow \quad K \gg T^{\frac{1}{2s}} \tag{14}$$

4.2 Upper Bound (Variance Control and Leakage Mitigation)

Estimating K parameters in OLS introduces in-sample variance inflation and potential "future leakage" bias. Although the α-mixing property of the weather process W_t with exponential decay prevents leakage from accumulating, the pure variance penalty scales as $K/\sqrt{T}$. To prevent variance inflation from inflating the denominator of the EL ratio, we require:

$$\frac{K}{\sqrt{T}} \to 0 \quad \implies \quad K \ll T^{\frac{1}{2}} \tag{15}$$

4.3 Growth rate of K

Combining these bounds yields the formal asymptotic growth condition:

$$T^{\frac{1}{2s}} \ll K \ll T^{\frac{1}{2}} \tag{16}$$

If we assume $m(W)$ is twice-differentiable ($s = 2$), the condition becomes $T^{1/4} \ll K \ll T^{1/2}$. In practice, setting $K \approx T^{1/3}$ satisfies these bounds, ensuring that both approximation bias and variance inflation vanish asymptotically.

5 The Formal Independence Assumption

To ensure the algebraic orthogonalization rigorously eliminates the nonparametric estimation bias, we must formally state the economic relationship between the financial market predictor and the exogenous weather shock. We impose the following two independence conditions on W_{t-1}:

Assumption 5.1 (Strict Exogeneity to the Market Shock) *The weather variable W_{t-1} is strictly exogenous to the financial market innovation U_t. Mathematically:*

$$\mathbb{E}[U_t \mid \mathcal{F}_{t-1}, W_{t-1}] = 0. \tag{17}$$

Remark. This condition says the realized local weather shock is not a proxy for the residual market innovation U_t after conditioning on predetermined market information. It still allows W_{t-1} to affect returns through the structural component $m(W_{t-1})$ and to be incorporated into future prices.

Assumption 5.2 (Contemporaneous Sieve Orthogonality Between Weather and Predictor Weight) *At the daily sampling frequency, realized local weather shocks are contemporaneously orthogonal to the predetermined persistent price predictor in the sieve directions used for profiling. For the EL weight $w(X_{t-1})$,*

$$\mathbb{E}[w(X_{t-1}) \mid W_{t-1}] = \mathbb{E}[w(X_{t-1})]. \tag{18}$$

Equivalently, for any centered nonconstant weather sieve vector $\boldsymbol{R}_K(W_{t-1})$,

$$\mathbb{E}\left[\boldsymbol{R}_K(W_{t-1})\{w(X_{t-1}) - \mathbb{E}w(X_{t-1})\}\right] = \mathbf{0}. \tag{19}$$

This is a timing restriction, not a dynamic exclusion: W_{t-1} may affect Y_t through $m(W_{t-1})$ and may affect subsequent returns and future prices.

Remark. This assumption is the population version of the sample orthogonality condition used in the proof, $\|T^{-1}\boldsymbol{R}'\tilde{\boldsymbol{w}}\| = O_p(\sqrt{K/T})$. It preserves the persistent predictor signal at the moment used for inference while allowing weather information to enter later price adjustment.

6 Wilks' Theorem for Mean Predictability

This section gives a formal Wilks theorem for the mean-predictability test. It is useful to write the nuisance step in null-imposed profile form. Let

$$\boldsymbol{Y} = (Y_1, \ldots, Y_T)', \qquad \boldsymbol{X}_{lag} = (X_0, \ldots, X_{T-1})',$$

and let $\boldsymbol{P}$ be the $T \times K$ matrix whose t-th row is $\boldsymbol{P}_K(W_{t-1})'$. Define

$$\boldsymbol{\Pi}_K = \boldsymbol{P}(\boldsymbol{P}'\boldsymbol{P})^{-1}\boldsymbol{P}', \qquad \boldsymbol{M}_K = \boldsymbol{I}_T - \boldsymbol{\Pi}_K.$$

For each candidate value β, define the null-imposed profile sieve coefficient and the corresponding residual by

$$\widehat{\boldsymbol{c}}_K(\beta) = (\boldsymbol{P}'\boldsymbol{P})^{-1}\boldsymbol{P}'(\boldsymbol{Y} - \beta \boldsymbol{X}_{lag}), \qquad \widehat{\boldsymbol{u}}(\beta) = \boldsymbol{M}_K(\boldsymbol{Y} - \beta \boldsymbol{X}_{lag}). \tag{20}$$

Let $w_t = w(X_{t-1})$, $\boldsymbol{w} = (w_1, \dots, w_T)'$, and define the sieve-orthogonalized weight

$$\boldsymbol{w}^c = \boldsymbol{M}_K \boldsymbol{w}, \qquad w_t^c = (\boldsymbol{w}^c)_t. \tag{21}$$

The scalar estimating equation is

$$\widehat{Z}_t(\beta) = \widehat{u}_t(\beta) w_t^c. \tag{22}$$

The empirical likelihood and its log-likelihood ratio are

$$L_T(\beta) = \sup\left\{\prod_{t=1}^T (T\pi_t) : \pi_t \geq 0, \quad \sum_{t=1}^T \pi_t = 1, \quad \sum_{t=1}^T \pi_t \widehat{Z}_t(\beta) = 0\right\}, \tag{23}$$

$$\ell_T(\beta) = -2\log L_T(\beta) = 2\sum_{t=1}^T \log\{1 + \widehat{\lambda}(\beta)\widehat{Z}_t(\beta)\}, \tag{24}$$

where $\widehat{\lambda}(\beta)$ solves

$$\sum_{t=1}^T \frac{\widehat{Z}_t(\beta)}{1 + \widehat{\lambda}(\beta)\widehat{Z}_t(\beta)} = 0. \tag{25}$$

For a vector $\boldsymbol{a} = (a_1, \dots, a_T)'$, write $\|\boldsymbol{a}\|_T^2 = T^{-1}\sum_{t=1}^T a_t^2$. Also write

$$\overline{w}_T = T^{-1}\sum_{t=1}^T w_t, \qquad \widetilde{w}_t = w_t - \overline{w}_T.$$

Assumption 1 (CCHT mean-score conditions)**.** *The predictor satisfies $X_t = \theta + \rho_T X_{t-1} + \varepsilon_t$, where ρ_T belongs to one of the persistence classes considered by Cai, Chen, Hong, and Tsvetanov (CCHT), including the stationary, mildly integrated, local-to-unity, integrated, and mildly explosive cases. With respect to the filtration*

$$\mathcal{H}_t = \sigma\{(z_s, \varepsilon_s) : s \leq t\} \vee \sigma\{W_s : s \leq t\},$$

$E(U_t \mid \mathcal{H}_{t-1}) = 0$, $\sup_t E|U_t|^{2+\delta} < \infty$ for some $\delta > 0$, and the conditional-variance and weight conditions used for the CCHT mean theorem hold. In particular, for

$$A_T = \frac{1}{\sqrt{T}}\sum_{t=1}^T U_t \widetilde{w}_t, \qquad V_T = \frac{1}{T}\sum_{t=1}^T U_t^2 \widetilde{w}_t^2,$$

there exists an a.s. positive, possibly random, variable η^2 such that

$$A_T \xrightarrow{\text{st}} MN(0, \eta^2), \qquad V_T \xrightarrow{p} \eta^2, \qquad \max_{1\leq t\leq T} |U_t \widetilde{w}_t| = o_p(\sqrt{T}). \tag{26}$$

The raw weight is uniformly bounded and satisfies the CCHT saturation condition.

Assumption 2 (Weather sieve orthogonality and projection)**.** *The first component of $\boldsymbol{P}_K(W)$ is one. After centering the remaining basis functions in sample, write the corresponding $T \times (K-1)$ matrix as $\boldsymbol{R}$, so that $\mathbf{1}_T'\boldsymbol{R} = \mathbf{0}$ and $\operatorname{span}(\boldsymbol{P}) = \operatorname{span}(\mathbf{1}_T, \boldsymbol{R})$. The primitive timing restriction is contemporaneous sieve orthogonality:*

$$\mathbb{E}\left[\boldsymbol{R}_K(W_{t-1})\{w(X_{t-1}) - \mathbb{E}w(X_{t-1})\}\right] = \mathbf{0},$$

where $\boldsymbol{R}_K(W_{t-1})$ denotes the population counterpart of the centered nonconstant sieve vector. This condition allows W_{t-1} to affect Y_t and future values of X; it only rules out contemporaneous projection of the predictor weight onto the weather sieve. The strict-exogeneity condition in Assumption 1 holds with contemporaneous weather information included in the conditioning set.

There are constants $0 < c < C < \infty$ such that, with probability tending to one,

$$c \leq \lambda_{\min}(T^{-1}\boldsymbol{R}'\boldsymbol{R}) \leq \lambda_{\max}(T^{-1}\boldsymbol{R}'\boldsymbol{R}) \leq C.$$

Let $\zeta_K = \max_{t\leq T} \|\boldsymbol{R}_t\|$, where $\boldsymbol{R}_t'$ is row t of $\boldsymbol{R}$. The sample analogue of the population orthogonality and the remaining increasing-dimensional score bounds are:

$$\left\|T^{-1}\boldsymbol{R}'\widetilde{\boldsymbol{w}}\right\| = O_p\left(\sqrt{K/T}\right), \tag{27}$$

$$\left\|T^{-1/2}\boldsymbol{R}'\boldsymbol{U}\right\| = O_p(\sqrt{K}), \tag{28}$$

$$\left\|T^{-1}\boldsymbol{P}'\boldsymbol{U}\right\| = O_p\left(\sqrt{K/T}\right), \tag{29}$$

where $\widetilde{\boldsymbol{w}} = (\widetilde{w}_1, \ldots, \widetilde{w}_T)'$ and $\boldsymbol{U} = (U_1, \ldots, U_T)'$. These bounds follow under the stated sieve orthogonality, exponential mixing of W_t, uniformly bounded basis moments, and the martingale moment condition in Assumption 1.

Assumption 3 (Approximation and growth rates)**.** *For some $\boldsymbol{c}_K$,*

$$m(W_{t-1}) = \boldsymbol{P}_K(W_{t-1})'\boldsymbol{c}_K + r_{K,t},$$

where

$$\|\boldsymbol{r}_K\|_T = O(a_K), \qquad \|\boldsymbol{r}_K\|_\infty = O(a_K), \qquad \sqrt{T}\, a_K \longrightarrow 0.$$

The sieve projector is stable in sup norm on the approximation residual, so that $\|\boldsymbol{M}_K\boldsymbol{r}_K\|_\infty = O(a_K)$. Moreover,

$$K/\sqrt{T} \longrightarrow 0, \qquad \zeta_K\sqrt{K/T} \longrightarrow 0. \tag{30}$$

For normalized spline bases, for which typically $\zeta_K \asymp \sqrt{K}$, the second condition in (30) is implied by $K/\sqrt{T} \to 0$. If $a_K \asymp K^{-s/d}$, a sufficient rate window is $T^{d/(2s)} \ll K \ll T^{1/2}$, which is nonempty when $s > d$.

Assumption 4 (Nondegeneracy and convex hull)**.** *There exists $\underline{\eta} > 0$ such that $P(\eta^2 \geq \underline{\eta}^2) = 1$. In addition,*

$$P\left\{\min_{1\leq t\leq T} \widehat{Z}_t(\beta_0) < 0 < \max_{1\leq t\leq T} \widehat{Z}_t(\beta_0)\right\} \longrightarrow 1.$$

Lemma 1 (Negligibility of the growing sieve projection)**.** *Under Assumptions 2–3,*

$$\|\boldsymbol{w}^c - \widetilde{\boldsymbol{w}}\|_T = O_p\left(\sqrt{K/T}\right), \tag{31}$$

$$\max_{t\leq T} |w_t^c - \widetilde{w}_t| = O_p\left(\zeta_K\sqrt{K/T}\right) = o_p(1), \tag{32}$$

$$\frac{1}{\sqrt{T}}\sum_{t=1}^{T} U_t(w_t^c - \widetilde{w}_t) = O_p(K/\sqrt{T}) = o_p(1), \tag{33}$$

$$\|\boldsymbol{\Pi}_K\boldsymbol{U}\|_T = O_p\left(\sqrt{K/T}\right), \tag{34}$$

$$\max_{t\leq T} |(\boldsymbol{\Pi}_K\boldsymbol{U})_t| = O_p\left(\zeta_K\sqrt{K/T}\right) = o_p(1). \tag{35}$$

Proof. Because the sieve contains a constant and $\boldsymbol{1}_T'\boldsymbol{R} = \boldsymbol{0}$, residualizing $\boldsymbol{w}$ on $\boldsymbol{P}$ is equivalent to residualizing $\widetilde{\boldsymbol{w}}$ on $\boldsymbol{R}$. Hence

$$\boldsymbol{w}^c = \widetilde{\boldsymbol{w}} - \boldsymbol{R}\widehat{\boldsymbol{\delta}}_K, \qquad \widehat{\boldsymbol{\delta}}_K = (\boldsymbol{R}'\boldsymbol{R})^{-1}\boldsymbol{R}'\widetilde{\boldsymbol{w}}.$$

The Gram-matrix condition and (27) imply $\|\widehat{\boldsymbol{\delta}}_K\| = O_p(\sqrt{K/T})$. Therefore,

$$\|\boldsymbol{R}\widehat{\boldsymbol{\delta}}_K\|_T^2 = \widehat{\boldsymbol{\delta}}_K'(T^{-1}\boldsymbol{R}'\boldsymbol{R})\widehat{\boldsymbol{\delta}}_K = O_p(K/T),$$

which proves (31); and

$$\max_t |\boldsymbol{R}_t'\widehat{\boldsymbol{\delta}}_K| \le \zeta_K \|\widehat{\boldsymbol{\delta}}_K\| = O_p\Big(\zeta_K\sqrt{K/T}\Big),$$

which proves (32). Moreover,

$$\frac{1}{\sqrt{T}}\boldsymbol{U}'(\boldsymbol{w}^c - \widetilde{\boldsymbol{w}}) = -\Big(T^{-1/2}\boldsymbol{R}'\boldsymbol{U}\Big)'\widehat{\boldsymbol{\delta}}_K = O_p(\sqrt{K})O_p\Big(\sqrt{K/T}\Big) = O_p(K/\sqrt{T}),$$

giving (33).

Finally, let $\widehat{\boldsymbol{a}}_U = (\boldsymbol{P}'\boldsymbol{P})^{-1}\boldsymbol{P}'\boldsymbol{U}$. The Gram condition and (29) imply $\|\widehat{\boldsymbol{a}}_U\| = O_p(\sqrt{K/T})$. The same quadratic-form and maximum-row arguments give (34) and (35). □

Lemma 2 (Asymptotic equivalence of the feasible and oracle scores)**.** *Under Assumptions 1–3, under $H_0 : \beta = \beta_0$,*

$$S_T := \frac{1}{\sqrt{T}}\sum_{t=1}^{T}\widehat{Z}_t(\beta_0) = \frac{1}{\sqrt{T}}\sum_{t=1}^{T}U_t\widetilde{w}_t + o_p(1), \tag{36}$$

$$\widehat{V}_T := \frac{1}{T}\sum_{t=1}^{T}\widehat{Z}_t(\beta_0)^2 = \frac{1}{T}\sum_{t=1}^{T}U_t^2\widetilde{w}_t^2 + o_p(1), \tag{37}$$

$$\max_{t\le T}|\widehat{Z}_t(\beta_0)| = o_p(\sqrt{T}). \tag{38}$$

Proof. Under the null,

$$\boldsymbol{Y} - \beta_0\boldsymbol{X}_{lag} = \boldsymbol{P}\boldsymbol{c}_K + \boldsymbol{r}_K + \boldsymbol{U}, \qquad \widehat{\boldsymbol{u}}(\beta_0) = \boldsymbol{M}_K(\boldsymbol{U} + \boldsymbol{r}_K).$$

Because $\boldsymbol{M}_K$ is symmetric and idempotent and $\boldsymbol{w}^c = \boldsymbol{M}_K\boldsymbol{w}$,

$$\begin{aligned} S_T &= T^{-1/2}\widehat{\boldsymbol{u}}(\beta_0)'\boldsymbol{w}^c \\ &= T^{-1/2}(\boldsymbol{U} + \boldsymbol{r}_K)'\boldsymbol{M}_K^2\boldsymbol{w} \\ &= T^{-1/2}(\boldsymbol{U} + \boldsymbol{r}_K)'\boldsymbol{w}^c. \end{aligned}$$

Lemma 1 gives $T^{-1/2}\boldsymbol{U}'\boldsymbol{w}^c = T^{-1/2}\boldsymbol{U}'\widetilde{\boldsymbol{w}} + o_p(1)$. Also, by Cauchy–Schwarz and the contraction property of an orthogonal projection,

$$\Big|T^{-1/2}\boldsymbol{r}_K'\boldsymbol{w}^c\Big| \le \sqrt{T}\,\|\boldsymbol{r}_K\|_T\|\boldsymbol{w}^c\|_T \le \sqrt{T}\,\|\boldsymbol{r}_K\|_T\|\boldsymbol{w}\|_T = o_p(1).$$

This proves (36).

For the second-moment result, define

$$e_{u,t} = \widehat{u}_t(\beta_0) - U_t, \qquad e_{w,t} = w_t^c - \widetilde{w}_t.$$

Then

$$\boldsymbol{e}_u = -\boldsymbol{\Pi}_K\boldsymbol{U} + \boldsymbol{M}_K\boldsymbol{r}_K,$$

so that, by Lemma 1 and projection contraction,

$$\|\boldsymbol{e}_u\|_T = O_p\Big(\sqrt{K/T} + a_K\Big) = o_p(1).$$

Moreover, $\max_t |e_{w,t}| = o_p(1)$, $\max_t |w_t^c| = O_p(1)$, and $T^{-1}\sum_t U_t^2 = O_p(1)$. Since

$$\widehat{Z}_t(\beta_0) - U_t\widetilde{w}_t = e_{u,t}w_t^c + U_te_{w,t},$$

it follows that

$$\frac{1}{T}\sum_{t=1}^{T}\{\widehat{Z}_t(\beta_0)-U_t\widetilde{w}_t\}^2 \le 2\max_t |w_t^c|^2\|\boldsymbol{e}_u\|_T^2 + 2\max_t |e_{w,t}|^2\frac{1}{T}\sum_{t=1}^{T}U_t^2$$
$$= o_p(1).$$

Equation (37) now follows from Cauchy–Schwarz.

Finally, $\max_t |U_t| = o_p(\sqrt{T})$ follows from the uniform $(2+\delta)$-moment bound. Together with (35), the sup-norm stability of $\boldsymbol{M}_K\boldsymbol{r}_K$, and $\max_t |w_t^c| = O_p(1)$, this proves (38). □

Theorem 1 (Wilks phenomenon for sieve-orthogonalized mean EL)**.** *Suppose Assumptions 1–4 hold. Then, under* $H_0 : \beta = \beta_0$*,*

$$\ell_T(\beta_0) \ \xrightarrow{d}\ \chi_1^2. \tag{39}$$

The conclusion holds for each predictor-persistence class covered by the CCHT mean theorem.

Proof. By Lemma 2 and Assumption 1,

$$S_T \ \xrightarrow{\text{st}}\ MN(0,\eta^2), \qquad \widehat{V}_T \ \xrightarrow{p} \eta^2.$$

Consequently,

$$\frac{S_T}{\sqrt{\widehat{V}_T}} \ \xrightarrow{\text{st}}\ N(0,1), \qquad \frac{S_T^2}{\widehat{V}_T} \ \xrightarrow{d} \chi_1^2. \tag{40}$$

It remains to connect the self-normalized statistic to empirical likelihood. Write $\widehat{Z}_t = \widehat{Z}_t(\beta_0)$ and $\widehat{\lambda} = \widehat{\lambda}(\beta_0)$. The convex-hull condition, $S_T = O_p(1)$, $\widehat{V}_T \to_p \eta^2 > 0$, and $\max_t |\widehat{Z}_t| = o_p(\sqrt{T})$ imply the standard scalar-EL bounds

$$\widehat{\lambda} = O_p(T^{-1/2}), \qquad \max_{t\le T} |\widehat{\lambda}\widehat{Z}_t| = o_p(1).$$

Expanding the multiplier equation (25) gives

$$0 = \sum_{t=1}^{T}\widehat{Z}_t - \widehat{\lambda}\sum_{t=1}^{T}\widehat{Z}_t^2 + o_p(\sqrt{T}),$$

and hence

$$\widehat{\lambda} = \frac{\sum_{t=1}^{T}\widehat{Z}_t}{\sum_{t=1}^{T}\widehat{Z}_t^2} + o_p(T^{-1/2}). \tag{41}$$

A second-order expansion of (24), together with $\max_t |\widehat{\lambda}\widehat{Z}_t| = o_p(1)$, yields

$$\begin{aligned}\ell_T(\beta_0) &= 2\widehat{\lambda}\sum_{t=1}^{T}\widehat{Z}_t - \widehat{\lambda}^2\sum_{t=1}^{T}\widehat{Z}_t^2 + o_p(1)\\ &= \frac{\left(T^{-1/2}\sum_{t=1}^{T}\widehat{Z}_t\right)^2}{T^{-1}\sum_{t=1}^{T}\widehat{Z}_t^2} + o_p(1)\\ &= \frac{S_T^2}{\widehat{V}_T} + o_p(1).\end{aligned}$$

The result follows from (40) and Slutsky's theorem. □

Corollary 1 (Test of no mean predictability)**.** *For* $H_0 : \beta = 0$*, reject at asymptotic level* α *whenever*

$$\ell_T(0) > \chi^2_{1,1-\alpha}.$$

Equivalently, the asymptotic p-value is $1 - F_{\chi_1^2}\{\ell_T(0)\}$*. A confidence set for* β *is obtained by inversion:*

$$\mathcal{C}_{1-\alpha} = \{\beta : \ell_T(\beta) \le \chi^2_{1,1-\alpha}\}.$$

Remark 1 (Relation to the unrestricted two-stage estimator)**.** *The profile construction* (20) *is the cleanest route to Theorem 1: at* β_0*, the fitted nuisance error is exactly* $\mathbf{\Pi}_K(\boldsymbol{U} + \boldsymbol{r}_K)$*, whose prediction norm is* $O_p(\sqrt{K/T} + a_K)$*. If one instead retains a separately estimated unrestricted coefficient* $\widehat{\mathbf{c}}$*, the same theorem continues to hold provided one proves the additional generated-nuisance condition*

$$\frac{1}{T}\sum_{t=1}^{T}\left\{\boldsymbol{P}_K(W_{t-1})'(\widehat{\boldsymbol{c}} - \boldsymbol{c}_K)\right\}^2 (w_t^c)^2 = o_p(1)$$

and the corresponding maximum-term bound. Those statements do not follow from the algebraic identity $\boldsymbol{P}'\boldsymbol{w}^c = \mathbf{0}$ *alone.*

Remark 2 (Why the same argument is not immediately a quantile theorem)**.** *For a quantile score* $\psi_\tau(u) = \tau - \mathbf{1}\{u < 0\}$*, the nuisance error enters nonlinearly:*

$$\psi_\tau(U_{t,\tau} + r_{K,t} - \boldsymbol{P}_K(W_{t-1})'(\widehat{\boldsymbol{c}}_\tau - \boldsymbol{c}_{K,\tau}))\,.$$

Therefore $\boldsymbol{P}'\boldsymbol{w}^c = \mathbf{0}$ *does not algebraically cancel the first-stage error. A first-order expansion instead produces a density-weighted term of the form*

$$\sum_{t=1}^{T} f_{t,\tau}(0)\boldsymbol{P}_K(W_{t-1})w_t^c,$$

which is not generally zero. A quantile extension would require either a restrictive condition making $f_{t,\tau}(0)$ *constant with respect to the predictor, or a new density-weighted orthogonalization and additional nonparametric-rate arguments. Thus Theorem 1 is a complete and natural endpoint for the mean paper, whereas a general quantile extension is a separate methodological problem.*

7 Conclusion

When researchers use standard predictive regressions, they are often forced to assume a constant intercept to maintain mathematical tractability. Consequently, all other exogenous macro-environmental factors are relegated to the error term. While this preserves the null hypothesis, it artificially inflates the residual variance, mathematically obliterating the test's power to detect true financial predictability. The proposed method uses the AR(1) specification only to describe the persistence class of the predictor. The empirical likelihood statistic itself does not estimate or include AR residuals. Instead, it profiles out the nonparametric nuisance $m(W)$ under each candidate β and residualizes the EL weight against the same sieve space. By doing so, the framework fully recovers the lost signal of the persistent predictor, offering a powerful tool for complex, real-world scenarios such as commodity pricing.